Estd:1980

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
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CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

Regulation: R20	**II / IV - B.Tech. I - Semester**								
INFORMATION TECHNOLOGY									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 BS 2101	Numerical Methods & Vector Calculus	BS	3	3	0	0	30	70	100
B20 IT 2101	Data Structures	PC	3	3	0	0	30	70	100
B20 IT 2102	Database Management Systems	PC	3	3	0	0	30	70	100
B20 IT 2103	Python Programming	PC	3	3	0	0	30	70	100
B20 IT 2104	Discrete Mathematics and Graph Theory	PC	3	3	0	0	30	70	100
B20 IT 2105	Data Structures Lab	PC	1.5	0	0	3	15	35	50
B20 IT 2106	Database Management Systems Lab	PC	1.5	0	0	3	15	35	50
B20 IT 2107	Python Programming Lab	PC	1.5	0	0	3	15	35	50
#SOC-I	Skill Oriented Course-I	SOC	2	0	0	4	--	50	50
B20 MC 2101	Environmental Science	MC	0	2	0	0	--	--	--
B20 MC 2103	English Proficiency	MC	0	2	0	0	--	--	--
		TOTAL	21.5	19	0	13	195	505	700

	Course Code	**Name of the Course**
#SOC-I	B20 IT 2108	Network Administration
	B20 IT 2109	spread Sheet Data Analysis

Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2101	**BS**	**3**	**--**	**--**	**3**	30	**70**	**3 Hrs.**

NUMERICAL METHODS & VECTOR CALCULUS

Common to CE, CSE, EEE&IT

Pre-requisites: Basic concepts of calculus.

Course Objectives: Students are expected to learn

1	Numerical methods to solve algebraic and transcendental equations and the concept of interpolation and its use for equally and unequally spaced data points,
2	Methods for numerical evaluation of integrals and for solving first order ODEs.
3	Concepts of double, triple integrals and its applications.
4	Concepts of Gradient, divergence, curl.
5	Vector integral theorems.

Course Out Comes: At the end of the course students will be able to

S. No	OUT COME	Knowledge Level
1	Determine a real root of an algebraic or transcendental equation. Fit an interpolation formula and perform interpolation for equally spaced and unequally spaced data.	K3
2	Evaluate numerically certain definite integrals. Solve a first order ordinary differential equation by Euler and RK methods.	K3
3	Evaluate double integrals and determine the areas.	K3
4	Evaluate triple integrals and determine the volumes.	K3
5	Find the gradient of a scalar function, divergence and curl of a vector function.	K3
6	Solve simple problems using vector integral theorems.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Solution of Algebraic and Transcendental Equations:** Introduction, Bisection method, Method of false position, Iteration method &Newton-Raphson method. **Interpolation:**Introduction, forward differences, backward differences, and Central differences. Differences of a polynomial, Newton's forward, and backward interpolation formulae. Interpolation with unequal intervals:Newton's divided difference and Lagrange interpolation formulae.
UNIT-II (10 Hrs)	**Numerical Integration and solution of Ordinary Differential equations:** Trapezoidal rule, Simpson's $1/3^{rd}$ rule. Solution of first order ordinary differential equations subjected to initial conditions by Taylor's method, Picard's method, Euler's method, Modified Euler's method and Fourth order Runge-Kutta method.

UNIT-III (12 Hrs)	**Multiple integrals:** Double and triple integrals, Change of order of integration. Change of variables, applications to finding Areas andVolumes.
UNIT-IV	**Vector differentiation:** Scalar and vector point functions, Vector Differentiation, Gradient, Directional derivative, Divergence, Curl, Scalar Potential.
UNIT-V (14Hrs)	**Vector Integration**: Line integral, Work done; Area, Surface and volume integrals, Vector integral theorems: Greens, Stokes, and Gauss Divergence theorems (without proof).

TEXTBOOKS:	
1.	Scope and Treatment as in “Higher Engineering Mathematics”, by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.
REFERENCE BOOKS:	
1.	Advanced Engineering Mathematics, by Erwin Kreyszig,Wiley
2.	Higher Engineering Mathematics, by B.V.Ramana, Tata McGraw Hill Company.
3.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
4.	Peter O’ Neil, Advanced Engineering Mathematics, Cengage.
5.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2101	PC	3	--	--	3	30	70	3 Hrs.

DATA STRUCTURES
(COMMON TO AIDS & IT)

Course Objectives:

1.	Introduce the fundamental concept of data structures and abstract data types
2.	Emphasize the importance of data structures in developing and implementing efficient algorithms
3.	Describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms
4.	

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	KL
1.	Illustrate different techniques for searching and sorting for given data.	K3
2.	Identify different parameters to analyze the performance of algorithms and implement linear data structures.	K3
3.	Design algorithms to perform operations with Non-Linear data structures.	K4

SYLLABUS

UNIT-I (10 Hrs)	**Data Structures -** Definition, Classification of Data Structures, Operations on Data Structures, Abstract Data Type (ADT), Preliminaries of algorithms. Time and Space complexity. **Searching** - Linear search, Binary search, Interpolation Search, Fibonacci search. **Sorting**- Insertion sort, Selection sort, Exchange (Bubble sort, quick sort), distribution (radix sort), merging (Merge sort) algorithms
UNIT-II (10 Hrs)	**Stacks:** Introduction to Stacks, Array Representation of Stacks, Operations on Stacks, Applications-Reversing list, Factorial Calculation, Infix to Postfix Conversion, Evaluating Postfix Expressions. **Queues:** Introduction to Queues, Representation of Queues-using Arrays, Implementation of Queues-using Arrays, Application of Queues-Circular Queues, Dequeues, Priority Queues, Multiple Queues.
UNIT-III (10 Hrs)	**Linked Lists:** Introduction, Singly linked list, Representation of Linked list in memory, Operations on Singly Linked list-Insertion, Deletion, Search and Traversal, Reversing Singly Linked list, Applications on Singly Linked list-Implementation of Stack and Queues, Polynomial Expression Representation, Addition and Multiplication, Sparse Matrix Representation using Linked List, Advantages and Disadvantages of Singly Linked list, Doubly Linked list-Insertion, Deletion, Circular Linked list-Insertion, Deletion.

UNIT-IV (8 Hrs)	**Trees:** Basic Terminology in Trees, Binary Trees-Properties, Representation of Binary Trees using Arrays and Linked lists. Binary Search Trees- Basic Concepts, BST Operations: Insertion, Deletion, Tree Traversals, Applications-Expression Trees, Heap Sort, Balanced Binary Trees- AVL Trees, Insertion, Deletion and Rotations.
UNIT-V (12 Hrs)	**Graphs:** Basic Concepts, Representations of Graphs-Adjacency Matrix and using Linked list, Graph Traversals (BFT & DFT), Applications- Minimum Spanning Tree Using Prims &Kruskals Algorithm, Dijkstra's shortest path, Transitive closure, Warshall's Algorithm.

Text Books:	
1.	Data Structures Using C. 2nd Edition.ReemaThareja, Oxford.
2.	Data Structures and algorithm analysis in C, 2nded, Mark Allen Weiss.
Reference Books:	
1.	Fundamentals of Data Structures in C, 2nd Edition, Horowitz, Sahni, Universities Press.
2.	Data Structures: A PseudoCode Approach, 2/e, Richard F.Gilberg, Behrouz A. Forouzon, Cengage.
3.	Data Structures with C, Seymour Lipschutz TMH

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2102	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

DATABASE MANAGEMENT SYSTEMS

(For IT)

Course Objectives:

1.	To introduce about database management systems
2.	To give a good formal foundation on the relational model of data and usage of Relational Algebra.
3.	To introduce the concepts of basic SQL as a universal Database language.
4.	To demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization
5.	To provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques

Course Outcomes: By the end of the course, student will be able to:

S. No	Outcome	Knowledge Level
1.	Understand fundamental concepts and architectures of database systems.	K2
2.	Develop database for an organization using E-R and Relational data models.	K3
3.	Apply knowledge of SQL to Create, Manipulate and Query databases.	K4
4.	Examine anomalies in database design and Apply Normalization concepts to refine the design.	K4
5.	Understand issues in transaction processing and data organization and efficient ways to address the same.	K2

SYLLABUS

UNIT-I (8 Hrs)	Introduction: Database system, Characteristics (Database Vs File System), Database Users (Actors on Scene, Workers Behind the Scene), Advantages of Database Systems, Database applications. Levels of abstraction offered by DBMS; Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Database system structure, Centralized and Client Server architecture (2-Tier and 3-Tier) for the database.
UNIT-II (10 Hrs)	**Entity Relationship Model**: Introduction, Representation of entities, Attributes, Entity Set, Relationship, Relationship set, Key Constraints, Participation Constraints, Weak Entity sets., Sub classes, super class, inheritance, specialization, generalization, aggregation using ER Diagrams. **Relational Model**: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, Referential Integrity constraints) and their importance. **BASIC SQL**: Simple Database Schema, Data Types, Table Definitions (Create, Alter), Different DML Operations (Insert, Delete, Update), Basic SQL Querying using Where Clause, Arithmetic Expressions in select clause, SQL Functions (Date and Time, Numeric, String Conversion).

UNIT-III (12 Hrs)	**Translating E-R diagrams to Relations**: Translating Entity Sets, Relationships, Weak Entity Sets, Class Hierarchies, Aggregation. **SQL**: Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), Relational set operations.
UNIT-IV (8 Hrs)	**Schema Refinement** (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency(1NF, 2NF and 3 NF), Concept of Surrogate key, Boyce-Codd normal form(BCNF), Lossless join and dependency preserving decomposition, Fourth normal form(4NF), Fifth Normal Form (5NF).
UNIT-V (12 Hrs)	**Transaction Concepts**: Transaction State, Transaction Properties, Concurrent Executions, Serializability, Anomalies due to Interleaved Execution, Precedence Graph to Test for Conflict Serializability, Failure Classification, Storage, Recovery and Atomicity, ARIES Recovery algorithm. **Indexing Techniques**: File Organizations and Indexing, Cluster Indexes, Primary and Secondary Indexes , Index data Structures: Hash Based Indexing, Tree base Indexing, Comparison of File Organizations, B+ Trees: Search, Insert, Delete algorithms.

Text Books:	
1.	Database Management Systems, 3/e, Raghurama Krishnan, Johannes Gehrke, TMH
2.	Database System Concepts,5/e, Silberschatz, Korth, TMH
Reference Books:	
1.	Introduction to Database Systems, 8/e C J Date, PEA.
2.	Database Management System, 6/e RamezElmasri, Shamkant B. Navathe, PEA
3.	Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2103	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

PYTHON PROGRAMMING
(Common to AIDS & IT)

Course Objectives:

1.	To learn about Python programming language syntax, semantics, and the runtime environment
2.	To be familiarized with universal computer programming concepts like data types, containers
3.	To be familiarized with general computer programming concepts like conditional execution, loops & functions
4.	To be familiarized with general coding techniques and object-oriented programming

Course Outcomes: At the end of this course, the student should be able to

S.No	Outcome	Knowledge Level
1.	Develop essential programming skills in computer programming concepts like data types, containers	**K4**
2.	Apply the basics of programming in the Python language	**K3**
3.	Solve coding tasks related conditional execution, loops	**K3**
4.	Solve coding tasks related to the fundamental notions and techniques used in object-oriented programming	**K3**
5	Implement the User defined exceptions and GUI application	**K3**

SYLLABUS

UNIT-I (10 Hrs)	**Introduction:** Introduction to Python, Program Development Cycle, Input, Processing, and Output, Displaying Output with the Print Function, Comments, Variables, Reading Input from the Keyboard, Performing Calculations, Operators. Type conversions, Expressions, More about Data Output. **Data Types, and Expression**: Strings Assignment, and Comment, Numeric Data Types and Character Sets, Using functions and Modules. **Decision Structures and Boolean Logic:** if, if-else, if-elif-else Statements, Nested Decision Structures, Comparing Strings, Logical Operators, Boolean Variables. Repetition Structures: Introduction, while loop, for loop, Calculating a Running Total, Input Validation Loops, Nested Loops.
UNIT-II (10 Hrs)	**Control Statement:** Definite iteration for Loop Formatting Text for output, Selection if and if else Statement, Conditional Iteration The While Loop **Strings and Text Files**: Accessing Character and Substring in Strings, Data Encryption, Strings and Number Systems, String Methods Text Files.

UNIT-III (10 Hrs)	**List and Dictionaries:** Lists, Defining Simple Functions, Dictionaries **Design with Function:** Functions as Abstraction Mechanisms, Problem Solving with Top Down Design, Design with Recursive Functions, Case Study Gathering Information from a File System, Managing a Program's Namespace, Higher Order Function. **Modules:** Modules, Standard Modules, Packages.
UNIT-IV (10 Hrs)	**File Operations:** Reading config files in python, Writing log files in python, Understanding read functions, read(), readline() and readlines(), Understanding write functions, write() and writelines(), Manipulating file pointer using seek, Programming using file operations **Object Oriented Programming:** Concept of class, object and instances, Constructor, class attributes and destructors, Real time use of class in live projects, Inheritance , overlapping and overloading operators, Adding and retrieving dynamic attributes of classes, Programming using Oops support **Design with Classes:** Objects and Classes, Data modeling Examples, Case Study An ATM, Structuring Classes with Inheritance and Polymorphism.
UNIT-V (10 Hrs)	**Errors and Exceptions:** Syntax Errors, Exceptions, Handling Exceptions, Raising Exceptions, User-defined Exceptions, Defining Clean-up Actions, Redefined Clean-up Actions. **Graphical User Interfaces:** The Behaviour of Terminal Based Programs and GUI -Based, Programs, Coding Simple GUI-Based Programs, Other Useful GUI Resources. **Programming:** Introduction to Programming Concepts with Scratch.

Text Books:	
1.	Fundamentals of Python First Programs, Kenneth. A. Lambert, Cengage.
2.	Python Programming: A Modern Approach, VamsiKurama, Pearson.

Reference Books:	
1.	Introduction to Python Programming, Gowrishankar.S, Veena A, CRC Press.
2.	Introduction to Programming Using Python, Y. Daniel Liang, Pearson

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2104	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

DISCRETE MATHEMATICS AND GRAPH THEORY

(For IT)

Course Objectives: Students are expected to	
1.	Understand propositional and predicate calculus.
2.	Know about concepts of counting techniques.
3.	Identify various types of relations and discuss their properties.
4.	Understand the concepts in Lattices and Boolean Algebra.
5.	Know about generating functions and methods of solving recurrence relations
6.	Have an idea on the concepts of Graph theory & Tree structures

	Course Outcomes: At the end of the course students will be able to	**KL**
1.	Write and verify the arguments for their validity using propositional and predicate logic.	**K3**
2.	Utilize different counting methodsin their fields of study.	**K3**
3.	Make use of various types of relations and their properties.	**K3**
4.	Identify different Lattices and Boolean expressions.	**K3**
5.	Formulate and solve the recurrence relations.	**K3**
6.	Utilize the concepts in graphs and trees.	**K3**

SYLLABUS

UNIT-I (12Hrs)	**Mathematical Logic:** Propositional Calculus: Statements and Notations, Connectives, Well-formed Formulae, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises. Predicate Calculus: Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.
UNIT-II (10Hrs)	**Combinatorics**: Basics of Counting, Permutations, Permutations with Repetitions, Circular Permutations, Restricted Permutations, Combinations, Restricted Combinations, Generating Functions of Permutations and Combinations, Binomial and Multinomial Theorems, Binomial and Multinomial Coefficients, Principle of Inclusion–Exclusion.
UNIT-III (14Hrs)	**Relations, Lattices & Boolean Algebra:** **Relations :**Definition of Relation, Properties of Binary Relations, Relation matrix and diagraph, Operations on Relations, Transitive Closure, Warshall's algorithm, Equivalence and Compatibility relations, Partial Ordering Relations, Hasse Diagrams. **Lattices & Boolean Algebra:** Lattices and their properties, different types of lattices, Boolean algebra- Boolean expressions, truth tables and karnaugh maps

UNIT-IV (10 Hrs)	**Recurrence Relations**: Generating Functions, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations
UNIT-V (12Hrs)	**Graph Theory**: Basic Concepts of Graphs, Sub graphs, Isomorphism of Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs, Multigraphs, Bipartite graphs, Planar Graphs, Euler's Formula. **Trees**: Definition of Tree, properties of Trees, Different tree structures, Binary trees, Spanning trees, Minimal Spanning Trees, Kruskal's and Prim's Algorithms.

Text Books:

1.	Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2.	Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India

Reference Books:

1.	Elements of Discrete Mathematics-A Computer Oriented Approach, C. L. Liu and D.P. Mahopatra, 3rd Edition, Tata McGraw Hill.
2.	Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
3.	Discrete Mathematical Structures, BernandKolman, Robert C. Busby, Sharon Cutler Ross, PHI.
4.	Discrete Mathematics, S. K. Chakraborthy and B.K. Sarkar, Oxford, 2011.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2105	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

DATA STRUCTURES LAB

(COMMON TO AIDS & IT)

Course Objectives:

1.	Demonstrate the different data structures implementation.

Course Outcomes: At the end of the course, the students will be able to:

S.No	Outcome	Knowledge Level
1.	Use basic data structures such as arrays and linked list.	K3
2.	Programs to demonstrate fundamental algorithmic problems including Tree Traversals, Graph traversals, and shortest paths.	K3
3.	Use various searching and sorting algorithms.	K3

LIST OF EXPERIMENTS

Exercise -1 (Searching)

Write C program that use both recursive and non-recursive functions to perform Linear search for a Key value in a given list.

b) Write C program that use both recursive and non-recursive functions to perform Binary search for a Key value in a given list.

Exercise – 2 (Sorting-I)

a) Write C program that implement Bubble sort, to sort a given list of integers in ascending order

b) Write C program that implement Quick sort, to sort a given list of integers in ascending order

c) Write C program that implement Insertion sort, to sort a given list of integers in ascending order

Exercise -3 (Sorting-II)

a) Write C program that implement radix sort, to sort a given list of integers in ascending order

b) Write C program that implement merge sort, to sort a given list of integers in ascending order

Exercise -4 (Stack)

a) Write C program that implement stack (its operations) using arrays

b Write a C program that uses Stack operations to evaluate postfix expression

Exercise -5(Queue)

a) Write C program that implement Queue (its operations) using arrays.

b) Write C program that implement Circular Queue (its operations) using arrays

Exercise -6 (Singly Linked List)

a) Write a C program that uses functions to create a singly linked list

b) Write a C program that uses functions to perform insertion operation on a singly linked list

c) Write a C program that uses functions to perform deletion operation on a singly linked list

d) Write a C program to reverse elements of a single linked list.

e) Write C program that implement stack (its operations) using Linked list.

f) Write C program that implement Queue (its operations) using Linked list.

Exercise -7 (Binary Search Tree) a) Write a C program to Create a BST b) Write a C program to insert a node into a BST. c) Write a C program to delete a node from a BST. d) Write a recursive C program for traversing a binary tree in preorder, inorder and postorder.
Text Books:
1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.
Reference Books:
Data Structures using C by Aaron M. Tenenbaum,Y.Langsam and M.J. Augenstein, Pearson Education, 2009.
Data Structures with C by Seymour lipschutz, Schaum Outline series, 2010.
Data Structures using C by R. KrishnaMoorthy G. IndiraniKumaravel, TMH, New Delhi,2008.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2106	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

DATABASE MANAGEMENT SYSTEMS LAB
(For IT)

Course Objectives:

1	Populate and query a database using SQL DDL/DML Commands
2	Declare and Enforce integrity constraints on a database
3	Writing Queries using advanced concepts of SQL
4	Programming PL/SQL including procedures, functions, cursors and triggers

Course Outcomes

S.No	Outcome	Knowledge Level
1	Utilize SQL to execute queries for creating database and performing data manipulation operations.	K4
2	Apply Queries using Advanced Concepts of SQL	K4
3	Build PL/SQL programs including stored procedures, functions and cursors.	K4
4	Create Triggers to initiate system responses for user actions.	K4

LIST OF EXPERIMENTS

1.	Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2.	Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3.	Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and Dropping of Views.
4.	Queries using Conversion functions (to_char, to_number and to_date), String Functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), Date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.	Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found) Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6.	Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions
7.	Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE- APPLICATION ERROR.
8.	Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES
9.	Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.

10.	Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables
11.	Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
12.	Create a table and perform the search operation on table using indexing and non-indexing techniques

Text Books:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle”, PHI, 2007
3. Rick F Vander Lans, “Introduction to SQL”, Fourth Edition, Pearson Education, 2007

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2107	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

PYTHON PROGRAMMING LAB
(Common to AIDS & IT)

Course Objectives: The student who successfully completes this course will have:

1	To acquire programming skills in core Python.
2	To acquire Object Oriented Skills in Python
3	To develop the skill of designing Graphical user Interfaces in Python
4	To develop the ability to write database applications in Python

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	KL
1	Write, Test and Debug Python Programs	K4
2	Use Conditionals and Loops for Python Programs	K3
3	Use functions and represent Compound data using Lists, Tuples and Dictionaries	K3
4	Use various applications using python	K3

SYLLABUS

1	Write a program that asks the user to enter three numbers (use three separate input statements). Create variables called total and average that hold the sum and average of the three numbers and print out the values of total and average.
2	Write a program that uses a *for* loop to print the numbers 8, 11, 14, 17, 20, .., 83, 86, 89.
3	Write a program that asks the user for their name and how many times to print it. The program should print out the user's name the specified number of times.
4	Use a *for*loop to print a triangle like the one below. Allow the user to specify how high the triangle should be. * ** *** ****
5	Write a program that asks the user to enter a word and prints out whether that word contains any vowels.
6	Write a program that asks the user to enter two strings of the same length. The program should then check to see if the strings are of the same length. If they are not, the program should print an appropriate message and exit. If they are of the same length, the program should alternate the characters of the two strings. For example, if the user enters *abcde*and *ABCDE* the program should print out *AaBbCcDdEe*.
7	Write a program that asks the user for a large integer and inserts commas into it according to the standard American convention for commas in large numbers. For instance, if the user enters 1000000, the output should be 1,000,000.
8	Write a program that generates a list of 20 random numbers between 1 and 100. (a) Print the list.

	(b) Print the average of the elements in the list. (c) Print the largest and smallest values in the list. (d) Print the second largest and second smallest entries in the list (e) Print how many even numbers are in the list.
9	Write a function called *sum_digits*that is given an integer num and returns the sum of the digits of num.
10	Write a function called *number_of_factors*that takes an integer and returns how many factors the number has.
11	Write a function called primes that is given a number n and returns a list of the first n primes. Let the default value of n be 100.
12	Write a function called merge that takes two already sorted lists of possibly different lengths, and merges them into a single sorted list. (a) Do this using the sort method. (b) Do this without using the sort method.
13	Write a program that asks the user for a word and finds all the smaller words that can be made from the letters of that word. The number of occurrences of a letter in a smaller word can't exceed the number of occurrences of the letter in the user's word.
14	Write a class called Product. The class should have fields called name, amount, and price, holding the product's name, the number of items of that product in stock, and the regular price of the product. There should be a method *get_price*that receives the number of items to be bought and returns a the cost of buying that many items, where the regular price is charged for orders of less than 10 items, a 10% discount is applied for orders of between 10 and 99 items, and a 20% discount is applied for orders of 100 or more items. There should also be a method called *make_purchase*that receives the number of items to be bought and decreases amount by that much.
15	Write a class called Time whose only field is a time in seconds. It should have a method called *convert_to_minutes*that returns a string of minutes and seconds formatted as in the following example: if seconds is 230, the method should return '5:50'. It should also have a method called *convert_to_hours*that returns a string of hours, minutes, and seconds formatted analogously to the previous method.
16	Write a Python class to implement pow(x, n).
17	Write a Python class to reverse a string word by word.
18	Write a program that opens a file dialog that allows you to select a text file. The program then displays the contents of the file in a textbox.
19	Write a program to demonstrate Try/except/else.
20	Write a program to demonstrate try/finally and with/as.
Reference Books:	
1.	Introduction to Python Programming, Gowrishankar.S, Veena A, CRC Press.
2.	Programming and Problem Solving with Python, Ashok NamdevKamthane, Amit Ashok Kamthane, TMH, 2019.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2108	**SOC**	**-**	**--**	**4**	**2**	**--**	**50**	**3 Hrs.**

NETWORK ADMINISTRATION

(Skill Oriented Course-I)

(Common to AIDS ,CSBS & IT)

Course Objectives: On completing this course student will be able to

1	Install different Operating Systems, antivirus and components
2	Install and configure windows server.
3	Install and configure different networking protocols and network tools.

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	KL
1	Demonstrate installation and configuration of Operating systems.	K2
2	Demonstrate installation and configuration of DNS, DHCP, set up adhoc networks.	K2
3	Demonstrate different network related tools and applications.	K2

SYLLABUS

1	Install MS Windows from the CD/DVD. We can try to install version from 10 to XP (8.1 recommended). Installation should include users, administrators, device configuration, patches and updates, antivir system, necessary environment configuration (i.e. NumLock, hiding of extension settings, etc.).
2	HW installation. Changing hard drives, preparing cables, setting of a separated network (course on real computers, hard drives dedicated for the course allows hardware sharing).
3	Install selected Linux distribution (Debian or Slackware recommended). Settings, as for MS Windows.
4	Text mode console commands. Cron, scripts.
5	Novell Netware administration, if anyone interested. Novell is not used for many years, but is useful to describe, how the server can work
6	Work with server - SQL, Apache, FTP/SFTP, mail. Mail robots.
7	Server as a firewall, ip sharing (NAT, DNAT, masquerade). Server behind the firewall.
8	Installation & configuration of Windows professional.
9	Installation & configuration of Windows server.
10	Installing and Configuring Terminal Services
11	Installing DNS. Implementing DNS in windows networks.
12	Installing and configuring DHCP.
13	Configuring & Implementing routing services
14	Configuration and setup of adhoc network and infrastructure network.
15	NET tools, Deployment of NETTOOLS.
16	Tracing of email origin using email trace pro utility.
17	Use of key logges and anti key logger to secure your system

Reference Books:

1	Mastering Windows 2000 Server by Mark Minasi, Sybex, ISBN: 0-7821-2774-6
2	Computer Networks- Andrew S Tanenbaum, 4th edition, Pearson Education

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2109	**SOC**	**-**	**--**	**4**	**2**	**--**	**50**	**3 Hrs.**

SPREAD SHEET DATA ANALYSIS
(Skill Oriented Course-I)
(Common to AIDS ,CSBS & IT)

Course Objectives: On completing this course student will be able to

1	To develop basic knowledge in Excel
2	To expose the various functions in Excel
3	To extend the skill to use data visualization
4	To analyze the real time datasets
5	To develop Pivot tables and VLOOKUP functions

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1	Describe common Excel functionality and features used for data science	K2
2	Analyze and construct the data Visualization	K3
3	Configure the programming environment	K2
4	Analyze real time data set	K3
5	Implement Pivot tables and VLOOKUP functions	K3

SYLLABUS

List of experiments

1	Study of basic functions in Excel
2	Working with Range Names and Tables
3	Cleaning Data with Text Functions.
4	Working with VLOOKUP functions and Pivot tables.
5	Demonstration of data Visualization
6	Importing data from external source into excel
7	Creating a data model
8	Exploring data with Pivot tables
9	Create a dash board for a given requirement.
10	Implement a data analytics for the real time data set.

Reference Books:

1.	Julio Cesar Rodriques Martino, " Hands – on Machine learning with Microsoft Excel", Packt Publication 2019
2.	Paul McFedries, "Excel data analysis for dunnies", john wiley and sone 2019.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2101	**MC**	**3**	**--**	**--**	**--**	**--**	**--**	**--**

ENVIRONMENTAL SCIENCE

(Common to CSE, ECE & IT)

Course Objectives: The objectives of the course are to impart:

1.	Overall understanding of the natural resources.
2.	Basic understanding of the ecosystem and its diversity.
3.	Acquaintance on various environmental challenges induced due to unplanned anthropogenic activities.
4.	An understanding of the environmental impact of developmental activities.
5.	Awareness on the social issues, environmental legislation and global treaties.

SYLLABUS

UNIT-I (8 Hrs)	**Multidisciplinary nature of Environmental Studies:** Definition, Scope and Importance – Sustainability: Stockholm and Rio Summit–Global Environmental Challenges: Global warming and climate change, acid rains, ozone layer depletion, population growth and explosion, effects;. Role of information technology in environment and human health. **Ecosystems:** Concept of an ecosystem. - Structure and function of an ecosystem; Producers, consumers and decomposers. - Energy flow in the ecosystem - Ecological succession. - Food chains, food webs and ecological pyramids; Introduction, types, characteristic features, structure and function of Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems.
UNIT-II (8 Hrs)	**Natural Resources: Natural resources and associated problems.** Forest resources: Use and over – exploitation, deforestation – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources: Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources. Food resources**:** World food problems, changes caused by non-agriculture activities-effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources. Land resources: Land as a resource, land degradation, Wasteland reclamation, man induced landslides, soil erosion and desertification; Role of an individual in conservation of natural resources; Equitable use of resources for sustainable lifestyles.
UNIT-III (8 Hrs)	**Biodiversity and its conservation:** Definition: genetic, species and ecosystem diversity-classification - Value of biodiversity: consumptive use, productive use, social-Biodiversity at national and local levels. India as a mega-diversity nation - Hot-sports of biodiversity - Threats to biodiversity: habitat loss, man-wildlife conflicts. - Endangered and endemic species of India – Conservation of biodiversity: conservation ofbiodiversity.

UNIT-IV (8 Hrs)	**Environmental Pollution:** Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Noise pollution, Nuclear hazards. Role of an individual in prevention of pollution. - Pollution case studies, Sustainable Life Studies. Impact of Fire Crackers on Men and his well being. **Solid Waste Management:** Sources, Classification, effects and control measures of urban and industrial solid wastes. Consumerism and waste products, Biomedical, Hazardous and e – waste management.
UNIT-V (8 Hrs)	**Social Issues and the Environment:** Urban problems related to energy -Water conservation, rain water harvesting-Resettlement and rehabilitation of people; its problems and concerns. Environmental ethics: Issues and possible solutions. Environmental Protection Act -Air (Prevention and Control of Pollution) Act. –Water (Prevention and control of Pollution) Act -Wildlife Protection Act -Forest Conservation Act-Issues involved in enforcement of environmental legislation.-Public awareness.
UNIT-VI (8 Hrs)	**Environmental Management**: Impact Assessment and its significance various stages of EIA, preparation of EMP and EIS, Environmental audit. Ecotourism, Green Campus – Green business and Greenpolitics. **The student should Visit an Industry / Ecosystem and submit a report individually on any issues related to Environmental Studies course and make a power point presentation.**

Text Books:	
1.	Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers,Vijayawada Rani; Pearson Education, Chennai
2.	Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford UniversityPress.
3.	Environmental Studies, P. N. Palanisamy, P. Manikandan, A.Geetha, andK.Manjula

Reference Books:	
1.	Text Book of Environmental Studies, Deeshita Dave & P. UdayaBhaskar, CengageLearning.
2.	A Textbook of Environmental Studies, Shaashi Chawla, TMH, NewDelhi
3.	Environmental Studies, Benny Joseph, Tata McGraw Hill Co, NewDelhi
4.	Perspectives in Environment Studies, AnubhaKaushik, C P Kaushik, New AgeInternational Publishers,2014

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2103	**MC**	**2**	**--**	**--**	**--**	**--**	**--**	**--**

ENGLISH PROFICIENCY

(Common to CSE, ECE & IT)

Course Objectives: The students will be able to

1.	Communicate their ideas and views effectively
2.	Practice language skills and improve their language competency.
3.	Know and perform well in real life contexts
4.	Identify and examine their self-attributes which require improvementand motivation.
5.	Build confidence and overcome their inhibitions, stage freight, nervousness etc.,
6.	Improve their reading skills.

Course Outcomes: The students will

S.No	Outcome	Knowledge Level
1.	Improve speaking skills.	K3
2.	Enhance their listening capabilities	K3
3.	Learn and practice the skills of composition writing.	K3
4.	Enhance their reading and understanding of different texts.	K3
5.	Improve their communication both in formal and informalcontexts.	K3
6.	Be confident in presentation skills.	K3

SYLLABUS

UNIT-I	**Listening Skills** Types of listening Hearing and Listening Listening as a receptive skill
UNIT-II	**Speaking Skills** Presentation skills Describing event/place/thing Extempore Debate Group Discussion
UNIT-III	**Reading Skills** Types of Reading (Intensive and Extensive reading, Skimming, Scanning) Reading/Summarizing News Paper Articles
UNIT-IV	**Writing Skills** Essay Writing (Argumentative, Analytical and Descriptive) E-Mail Writing Business Letters Resume Writing

UNIT-V	**Integrated Language Skills** Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking	
Reference Books:		
1.	Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP	
2.	English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009	
3.	Enriching Speaking and Writing Skills, Orient Blackswan Publishers.	
4.	The Oxford Guide to Writing and Speaking by John Seely OUP.	
5.	Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw hill.	
6.	Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers	
Note: Internal Assessment is carried out throughout the semester.		

Regulation: R20	II / IV - B.Tech. II - Semester								
INFORMATION TECHNOLOGY									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 HS 2201	Managerial Economics and Financial Accountancy	HS	3	3	0	0	30	70	100
* B20 BS 2202	Statistics with R	BS	3	2	0	0	30	70	100
				0	0	2	15	35	50
B20 IT 2201	Computer Organization	PC	3	3	0	0	30	70	100
B20 IT 2202	Java Programming	ES	3	3	0	0	30	70	100
B20 IT 2203	Principles of Software Engineering	PC	3	3	0	0	30	70	100
B20 IT 2204	Unified Modeling Language (UML) Lab	PC	1.5	0	0	3	15	35	50
B20 IT 2205	Free Open-Source Software Lab	PC	1.5	0	0	3	15	35	50
B20 IT 2206	Java Programming Lab	ES	1.5	0	0	3	15	35	50
#SOC-II	Skill Oriented Course-II	SOC	2	0	0	4	--	50	50
		TOTAL	**21.5**	**14**	**0**	**15**	**210**	**540**	**750**

	Course Code	**Name of the Course**
#SOC-II	B20 IT 2207	Animations
	B20 IT 2208	Web Design Using PHP

Note:*- Integrated course and its evaluation guide lines are mentioned in the Syllabus

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20HS2201	**HS**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY

(Common to CSE & IT)

Course Objectives: Students are expected to learn

1.	To Study Managerial Economics and Demand Analysis
2.	To familiarize about the Concepts of Cost and Break-Even Analysis.
3.	To understand the nature of markets and to know the Pricing Policies
4.	To learn about accounting cycle and preparation of Financial Statements.
5.	To know the concept of Capital and sources of raising and Depreciation

Course Outcomes: At the end of the course the student will be able to

S. No	Outcome	KL
1.	Equip oneself with the knowledge of estimating the Demand and demand elasticities for a product.	K2
2.	Have knowledge of Cost and its types and ability to calculate BEP	K3
3.	Understand the nature of different markets	K2
4.	Uunderstand Pricing Practices prevailing in today's business world	K2
5.	Prepare Financial Statements and know how to calculate Profit & Loss for a firm	K3
6.	Know Types of capital and their sources and know how to calculate Depreciation	K2

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Managerial Economics and demand Analysis:** **Managerial Economics:** Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics.**Demand Analysis:** Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of demand forecasting and its Methods.
UNIT-II (10 Hrs)	**Cost Analysis**: Importance of cost analysis, **Types of Cost-** Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; **Elements of costs** – Material, Labour, Expenses; **Methods of costing** - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. **Break-even analysis:** Determination of Breakeven point - Applications, Assumptions and Limitations of Break -even analysis (Theory only).

UNIT-III (10 Hrs)	**Introduction to Markets & Pricing Policies** **Market Structures:** Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. **Pricing:** Importance of pricing and its meaning;**Methods of Pricing**: **Cost Based** -Full cost, Mark-up, Marginal &Break-even; **Demand Based** - Penetrating, Skimming; **Competition Based-** Going rate, Sealed Bid, Discount; **Internet Pricing** - Flat-rate, Usage sensitive.
UNIT-IV (8 Hrs)	**Introduction to Financial Accounting:** Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (outlines only).
UNIT-V (12 Hrs)	**Capital & Depreciation:** Types of Capital - Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, medium term and Long term. **Depreciation -** Meaning, Importance and causes of depreciation; Methods of Depreciation- Straight line and diminishing balancing methods (Theory only)

Text Books:

1.	A R Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N.Appa Rao, Dr.P. Vijayakumar:Managerial Economics and Financial Analysis', Cengage Publications, New Delhi

Reference Books:

1.	Dr.B.Kuberudu& T.V. Ramana : Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K. Gupta & R.K. Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2202	BS	2	--	2	3	Theory: 30	Theory: 70	Theory: 3 Hrs.
						Lab:15	Lab:35	Lab: 3 Hrs.

STATISTICS with R

(For IT)

COURSE OBJECTIVES: Students are expected to

1	Have an idea of discrete and continuous random variables and fundamentals of R.
2	Learn the concept of probability distribution models and R functions for distribution models
3	Discuss sampling distribution, estimation and R functions for constructing confidence intervals
4	Know about the hypothesis testing for means and variance and related R functions
5	Develop a framework for testing of hypothesis in giving inferences about Population parameters.
6	Explain correlation and regression models and R functions for graphics

Course Out Comes: At the end of the course students will be able to

S. No	OUT COME	KL
1	Identify discrete and continuous random variables and data structures in R	**K2**
2	Apply discrete and continuous probability distributions to the given data and Execute R functions for probability distributions.	**K3**
3	Explain sampling distribution, estimation, and R functions for constructing confidence intervals	**K3**
4	Test the given hypothesis of population based on a sample data	**K3**
5	Describe built in R functions for sampling tests	**K3**
6	Apply the concepts of correlation and regression to the given statistical data. Make use of graphic functions to visualize the data.	**K3**

SYLLABUS

UNIT-I (12 Hrs.)	**Random Variables and Introduction to R** Random Variables- Discrete, Continuous random variables-Expectation, Variance, Moment Generating Function Introduction to R software – Vectors – Matrices – Arrays – Lists – Data frames – Basic arithmetic operations in R – Importing and exporting files in R.
UNIT-II (10 Hrs.)	**Probability Distributions** Discrete Probability distributions- Binomial Distribution, Poisson Distribution Continuous Probability Distributions- Normal Distribution- Gamma distribution R commands for computing probability distributions.
UNIT-III (12 Hrs.)	**Sampling Theory and Test of Hypothesis** Sampling – Central limit theorem (without proof) – Sampling distribution of means – point estimation – interval estimation Construction of confidence intervals using R.

UNIT-IV (12 Hrs.)	**Test of Significance** Hypothesis, types, One tailed, Two tailed, Errors in Sampling, Test for single and difference of proportions, t-tests-test for single mean and difference of means,F-test, Chi-square test for goodness of fit. R programming for t-test, F-test,Chi-square test.
UNIT-V (12 Hrs.)	**Correlation, Regression& Curve Fitting:** Correlation and regression, Regression by the method of least squares, Multiple linear Regression, Method of least squares, Fitting of Straight line and second-degree parabola. R programming for correlation and regression, Basic Graphics-plot, Creating Graphs for regressions.

EXPERIMENTS

1.	Write a R program to take input from the user (name and age) and display the values. Also print the version of R installation.
2.	Write a R program to create a sequence of numbers from 20 to 50 and find the mean of numbers from 20 to 60 and sum of numbers from 51 to 91.
3.	Write a R program to create three vectors a,b,c with 3 integers. Combine the three vectors to become a 3×3 matrix where each column represents a vector. Print the content of the matrix.
4.	Write a R program to find row and column index of maximum and minimum value in a given matrix.
5	Write a R program to combine three arrays so that the first row of the first array is followed by the first row of the second array and then first row of the third array
6	Write a R program to create an array using four given columns, three given rows, and two given tables and display the content of the array.
7	Write a R program to create a data frame from four given vectors.
8	Write a R program to find Sum, Mean and Product of a Vector, ignore element like NA or NaN.
9	Write a R program to create a list containing a vector, a matrix and a list and remove the second element.
10	Write a R program to merge two given lists into one list
11	Write a R program to create an ordered factor from data consisting of the names of months.
12	Plot the density and distribution functions for Normal approximation to the Binomial distribution
13	Take any dataset, Visualize Tables, charts and plots. Compute visualising Measures of Central Tendency, Variation, and Shape. Box plots, Pareto diagrams. Aslo, find the mean median standard deviation and quantiles of a set of observations.
14	Take any dataset. Calculate the correlation between two variables. Draw the scatter plots. Use the scatter plot to investigate the relationship between two variables
15	The sales of a company for each year are shown in the table below. x (year) 2015 2016 2017 2018 2019 y (sales in lakhs) 12 19 29 37 45 a) Find the least square regression line y = a x + b. b) Use the least squares regression line as a model to estimate the sales of the company in 2021
16	Find the least square regression line for the following set of data {(-1 , 0),(0 , 2),(1 , 4),(2 , 5) Plot the given points and the regression line in the same rectangular system of axes

TEXT BOOKS:	
1.	G. Jay Kerns, Introduction to Probability and Statistics Using R, First Edition ISBN: 978-0-557-24979-4. (Free e-book from R software website)
2.	Fundamentals of Mathematical Statistics by S. C. Gupta and V.K. Kapoor, Sultan Chand & Sons Publishers.
3	The Art of R Programming, Norman Matloff, No starch Press..
REFERENCE BOOKS:	
1.	Higher Engineering Mathematics, by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.
2.	Probability and statistics for Engineers, Miller and Freund, 7th edition, Prentice-Hall India.
3.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.
4.	R Cookbook, Paul Teetor,Oreilly.
5.	R in Action, Rob Kabacoff, Manning
6	R for Everyone, Lander, Pearson
WEB REFERENCE	
1.	http://www.swayam.gov.in
2.	https://www.tutorialspoint.com/r/
3.	http://www.stat.umn.edu/geyer/old/5101/rlook.html
4.	http://www.r-tutor.com/elementary-statistics
5.	https://onlinecourses.nptel.ac.in/noc16_ma03/preview

Evaluation guidelines for the integrated course:

The Student has to pass both theory and lab examinations separately in order to complete the Integrated Course. If the Student fails in either theory or lab, he/she has to reappear for both theory and lab in supplementary examinations. Student will be declared as pass only when he/she completes both theory and lab at the same time.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2201	**PC**	3	**--**	--	**3**	**30**	**70**	**3 Hrs.**

COMPUTER ORGANIZATION

(For IT)

Course Objectives: The course objectives of Computer Organization are to discuss and make student familiar with the

1.	Principles and the Implementation of Computer Arithmetic
2.	Operation of CPUs including RTL, ALU, Instruction Cycle and Busses
3.	Fundamentals of different Instruction Set Architectures and their relationship to the CPU Design
4.	Memory System and I/O Organization
5.	Principles of Operation of Multiprocessor Systems and Pipelining

Course Outcomes: By the end of the course, the student will

S.No	Outcome	Knowledge Level
1.	Illustrate the various data representations, notations, arithmetic algorithms and flow control for various instructions using micro operations in basic computer	K2
2.	Detailed understanding of architecture and functionality of central processing unit and various control units	K2
3.	Exemplify in a better way the I/O and memory organization	K3
4.	Illustrate concepts of parallel processing, pipelining and inter processor communication	K2

SYLLABUS

UNIT-I (10 Hrs)	**Basic Structure of Computers**: Basic Organization of Computers, Historical Perspective, Bus Structures, Data Representation: Data types, Complements, Fixed Point Representation. Floating – Point Representation. Other Binary Codes, Error Detection Codes. Computer Arithmetic: Addition and Subtraction, Multiplication Algorithms, Division Algorithms.
UNIT-II (10 Hrs)	**Register Transfer Language and Microoperations**: Register Transfer language. Register Transfer Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit. **Basic Computer Organization and Design**: Instruction Codes, Computer Register, Computer Instructions, Instruction Cycle, Memory –Reference Instructions. Input –Output and Interrupt, Complete Computer Description,
UNIT-III (10 Hrs)	**Central Processing Unit**: General Register Organization, STACK Organization. Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer. **Micro programmed Control**: Control Memory, Address Sequencing, Micro Program example, Design of Control Unit

UNIT-IV (8 Hrs)	**Memory Organization**: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory. Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupts, Direct Memory Access.
UNIT-V (12 Hrs)	Multi Processors: Introduction, Characteristics of Multiprocessors, Interconnection Structures, Inter Processor Arbitration. Pipeline: Parallel Processing, Pipelining, Instruction Pipeline,

Text Books:	
1.	Computer System Architecture, M. Morris Mano, Third Edition, Pearson, 2008.
2.	Computer Organization, Carl Hamacher, ZvonkoVranesic, SafwatZaky, 5/e, McGraw Hill, 2002.
Reference Books:	
1.	Computer Organization and Architecture, William Stallings, 6/e, Pearson, 2006.
2.	Structured Computer Organization, Andrew S. Tanenbaum, 4/e, Pearson, 2005.
3.	Fundamentals of Computer Organization and Design, Sivarama P. Dandamudi, Springer, 2006
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105163/
2.	http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2202	PC	3	--	--	3	30	70	3 Hrs.

JAVA PROGRAMMING

(COMMON TO AIDS &IT)

Course Objectives:

1.	To identify Java language components and how they work together in applications
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications
4.	To understand how to design applications with threads in Java
5	To understand how to use Java APIs for program development

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1.	Able to apply the concepts of Object-Oriented Programming & Java Programming Constructs	K3
2.	Able to understand the basic concepts of Java such as operators, classes, objects, and various keywords	K2
3.	Apply the concept of Inheritance, Interfaces and Overriding the methods	K3
4.	Able to Analyze the applications of Java using Multithreading, Exception handling	K3
5.	Able to Analyze & Design the concept of Event Handling and Abstract Window Toolkit	K4

SYLLABUS

UNIT-I (10 Hrs)	**Program Structure in Java:** Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style. **Data Types, Variables, and Operators :**Introduction, Data Types in Java, Declaration ofVariables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants,Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods,Attribute Final, Introduction to Operators, Precedence and Associativity of Operators,Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -)Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, BitwiseLogical Operators. **Control Statements:** Introduction, if Expression, Nested if Expressions, if–else Expressions,Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do–while Loop, for Loop, Nested for Loop, For–Each for Loop, Break Statement, Continue Statement.
UNIT-II (10 Hrs)	**Classes and Objects:** Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for ClassMembers, Accessing Private Members of Class, Constructor Methods for Class, OverloadedConstructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Valueand byReference, Keyword this.

	Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded ConstructorMethods, Class Objects as Parameters in Methods, Access Control, Recursive Methods,Nesting of Methods,Attributes Final and Static.
UNIT-III (10 Hrs)	**Arrays**: Introduction, Declaration and Initialization of Arrays, Storage of Array in ComputerMemory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array toAnother Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values inArrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, ThreedimensionalArrays, Arrays as Vectors. **Inheritance:** Introduction, Process of Inheritance, Types of Inheritances, Universal SuperClass-Object Class, Inhibiting Inheritance of Class Using Final, Access Control andInheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method andInheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces andInheritance. **Interfaces:** Introduction, Declaration of Interface, Implementation of Interface, MultipleInterfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, StaticMethods in Interface, Functional Interfaces, Annotations.
UNIT-IV (8 Hrs)	**Packages and Java Library:** Introduction, Defining Package, Importing Packages and Classesinto Programs, Access Control, Packages in Java SE:Java.lang Package, Java utilandTimePackages. **Exception Handling:** Introduction, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Custom Exceptions, Nested try and catch Blocks,ThrowsClause. **String Handling in Java**: Introduction, Class String handling Methods, Class String Buffer. **Multithreaded Programming:** Introduction, Thread Class, Main Thread- Creation of NewThreads, Thread States, Runnable Interface, Thread Priority-Synchronization.
UNIT-V (12 Hrs)	**GUI programming with Swing:** Introduction, limitations of AWT, MVC Architecture, containers. Understanding Layout Managers: Flow, Border, Grid, Card, GridBag. **Event Handling:** The Delegation event model-Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events, Adapter classes, Inner classes, Inner classes, Inner classes, Anonymous Inner classes. A Simple Swing Application. Exploring swing controls-JLabel, JText field, The Swing Buttons-JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList ,JCombo Box, Swing Menus, Dialogs. **Java Database Connectivity:** Introduction, JDBC Architecture,Establishing JDBC Database Connections.

Text Books:

1.	JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2.	The complete Reference Java, 8th edition, Herbert Schildt, TMH.

Reference Books:

1.	Introduction to java programming, 7th edition by Y Daniel Liang, Pearson
2.	Murach's Java Programming, Joel Murach

e-Resources:

1) https://nptel.ac.in/courses/106/105/106105191/
2) ps://www.w3schools.com/java/java_data_types.asp

SubjectCode	Category	L	T	P	C	I.M	E.M	Exam
B20IT2203	PC	3	--	--	3	30	70	3 Hrs.

PRINCIPLESOFSOFTWAREENGINEERING

(For IT)

CourseObjectives: Thiscourseisdesignedto:

1.	GiveexposuretophasesofSoftwareDevelopment,commonprocessmodelsincludingWaterfall,andtheUnifiedProcess,andhands-onexperiencewithelementsoftheagileprocess
2.	GiveexposuretoavarietyofSoftwareEngineeringpracticessuchasrequirementsanalysis andspecification,codeanalysis,codedebugging,testing,traceability,andversioncontrol
3.	GiveexposuretoSoftwareDesigntechniques

CourseOutcomes: Studentstakingthissubjectwillgainsoftwareengineeringskillsinthefollowing areas:

S.No	Outcome	Knowledge Level
1.	Applysoftware engineering concepts to illustrate various phases of a lifecycle and choose appropriate process model depending on the user requirements.	K3
2.	Perform requirements analysis and design UML diagrams for the requirements gathered.	K4
3.	Implement various processes used in all the phases of the product.	K4
4.	Testwhetheralltherequirementsspecifiedhavebeenachievedornot	K4

SYLLABUS

UNIT-I(10Hrs)	TheNatureofSoftware,TheUniqueNatureofWebApps,SoftwareEngineering,TheSoftware Process, Software Engineering Practice, Software Myths, How It All Starts. AGenericProcessModel,ProcessAssessmentandImprovement,PrescriptiveProcessModels,SpecializedProcessModels,TheUnifiedProcess,PersonalandTeamProcess Models,ProcessTechnology.
UNIT-II (10Hrs)	Agility, Agility and the Cost of Change, Agile Process, Extreme Programming (XP), OtherAgile Process Models, A Tool Set for the Agile Process, Software Engineering Knowledge,CorePrinciples,PrinciplesThatGuideEachFrameworkActivity,RequirementsEngineering,EstablishingtheGroundwork,ElicitingRequirements,DevelopingUseCases, BuildingtheRequirementsModel,NegotiatingRequirements,ValidatingRequirements.
UNIT-III(10Hrs)	Requirements Analysis, Scenario-Based Modeling, UML Models That Supplement the UseCase, Data Modeling Concepts, Class-Based Modeling, Requirements Modeling Strategies, Flow Oriented Modeling, Creating a Behavioral Model, Patterns for Requirements Modeling, RequirementsModelingforWebApps.
UNIT-IV (8Hrs)	Design within the Context of Software Engineering, The Design Process, Design Concepts,TheDesignModel,SoftwareArchitecture,ArchitecturalGenres,ArchitecturalStyles,AssessingAlternativeArchitecturalDesigns,ArchitecturalMappingUsingDataFlow,WhatIsaComponent?,DesigningClass-BasedComponents,ConductingComponent-LevelDesign,Component-LevelDesignforWebApps,DesigningTraditionalComponents, Component-Based Development.

UNIT-V (12Hrs)	The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface DesignSteps,WebAppInterfaceDesign,DesignEvaluation,ElementsofSoftwareQualityAssurance, SQA Tasks, Goals & Metrics, Statistical SQA, Software Reliability, A Strategic Approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software,TestStrategiesforObject-OrientedSoftware,TestStrategiesforWebApps,ValidationTesting,SystemTesting,TheArtofDebugging,SoftwareTesting Fundamentals, Internal and External Views of Testing, White-Box Testing, Basis PathTesting

Text Books:	
1.	Software Engineering a practitioner's approach, Roger S. Pressman, Seventh Edition, McGraw Hill Higher Education.
2.	Software Engineering, Ian Sommerville, Ninth Edition, Pearson.
Reference Books:	
1.	Software Engineering, A Precise Approach, PankajJalote, Wiley India, 2010.
2.	Software Engineering, UgrasenSuman, Cengage.
e-Resources:	
1.	NPTEL :: Computer Science and Engineering - NOC:Software Engineering

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2204	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3Hrs.**

UNIFIED MODELING LANGUAGE (UML) LAB

(For IT)

Course Objectives: This course is designed to:

1.	Know the practical issues of the different object oriented analysis and design concepts.
2.	Inculcate the art of object oriented software analysis and design.
3.	Apply forward and reverse engineering of a software system.
4	Carry out the analysis and design of a system in an object oriented way.

Course Outcomes: Students will be able to:

S.No	Outcome	KL
1.	Practice the syntax of different UML diagrams.	K3
2.	Design UML diagrams that model both domain model and design model of a software system.	K4
3.	Implement the source code and develop significant projects.	K4
4.	Test and document the software.	K5

LIST OF EXPERIMENTS

Experiment 1:
Familiarization with Rational Rose or Umbrella environment.

Experiment 2:
Identify and analyze events. b) Identify usecases.c) Develop event table.

Experiment 3:
Identify and analyze domain classes.
Represent usecases and a domain class diagram using Rational Rose.
Develop CRUD matrix to represent relationships between usecases and problem domain classes.

Experiment 4:
Develop usecase diagrams.
Develop elaborate usecase description and scenarios.
Develop prototypes (without functionality).

Experiment 5:
Develop system sequence diagrams and high-level sequence diagrams for each usecase.
Identify MVC classes/objects for each usecase.
Develop detailed sequence diagrams/communication diagrams for each usecase showing interacting among all the three-layer objects.

Experiment 6:
Develop detailed design class model (use GRASP pattern for responsibility assignment)
Develop three-layer package diagrams for each case study.

Experiment 7:
Develop usecase packages.
Develop component diagrams.
Refine domain class model by showing all the associations among classes.

Experiment 8: Develop sample diagrams for other UML diagrams – state chart diagrams, activity diagrams and deployment diagrams.
Reference Books:
1. Project-based software engineering: An object oriented approach, Evelyn Stiller, Cathie LeBlanc and Pearson Education
2. Visual Modeling with Rational Rose 2002 and UML, Terry Quatrini, Pearson Education

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2205	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3Hrs.**

FREE OPEN-SOURCE SOFTWARE LAB

(For IT)

Course Objectives:

1.	Provide introduction to UNIX operating system and its File System.
2.	develop the ability to formulate Regular Expressions and use them for Pattern Matching
3.	Gain an understanding of important aspects related to the Shell and the Process.
4.	Provide a comprehensive introduction to Shell Programming, Services and Utilities.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Demonstrate UNIX commands for file handling and process control	K3
2.	Construct regular expressions for pattern matching and apply them to various filters for a specific task	K3
3.	Analyze a given problem and apply requisite facets of shell programming in order to devise a shell script to solve the problem	K3

LIST OF EXPERIMENTS

1. Session-1
a) Log into the system
b) Use vi editor to create a file called myfile.txt which contains some text.
c) Correct typing errors during creation.
d)Save the file
e) logout of the system
Session-2
a) Log into the system
b) open the file created in session 1
c)Add some text
d)Change some text
e) Delete some text
f) Save the Changes
g) Logout of the system

2.a) Log into the system
b) Use the cat command to create a file containing the following data. Call it mytable use tabs to separate the fields.
1425 Ravi 15.65
4320 Ramu26.27
6830 Sita 36.15
1450 Raju 21.86
c) Use the cat command to display the file, mytable
d)Use the vi command to correct any errors in the file, mytable.
e)Use the sort command to sort the file my table according to the first field. Call the sorted file my table (same name)

f)Print the file my table g)Use the cut and paste commands to swap fields 2 and 3 of my table. Call it my table (same name) h)Print the new file, mytable i)Logout of the system.
3.A) a) Login to the system b) Use the appropriate command to determine your login shell c) Use the /etc/passwd file to verify the result of step b. d) Use the who command and redirect the result to a file called myfile1. Use the more command to see the contents of myfile1. e) Use the date and who commands in sequence (in one line) such that the output of date will display on the screen and the output of who will be redirected to a file called myfile2. Use the more command to check the contents of myfile2. B)Write a sed command that deletes the first character in each line in a file. b)Write a sed command that deletes the character before the last character in each line in a file. c)Write a sed command that swaps the first and second words in each line in a file.
4. a)Pipe your /etc/passwd file to awk, and print out the home directory of each user. b)Develop an interactive grep script that asks for a word and a file name and then tells how many lines contain that word. c)Repeat d)Part using awk
5. a)Write a shell script that takes a command –line argument and reports on whether it is directory, a file, or something else b)Write a shell script that accepts one or more file name as arguments and converts all of them to uppercase, provided they exist in the current directory. c)Write a shell script that determines the period for which a specified user is working on the system.
6. a)Write a shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers. b)Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
7. a)Write a shell script that computes the gross salary of a employee according to the following rules: i)If basic salary is < 1500 then HRA =10% of the basic and DA =90% of the basic. ii)If basic salary is >=1500 then HRA =Rs500 and DA=98% of the basic The basic salary is entered interactively through the key board. b)Write a shell script that accepts two integers as its arguments and computers the value of first number raised to the power of the second number.
8. a)Write an interactive file-handling shell program. Let it offer the user the choice of copying, removing, renaming, or linking files. Once the user has made a choice, have the program ask the user for the necessary information, such as the file name, new name and so on. b)Write shell script that takes a login name as command – line argument and reports when that person logs in c)Write a shell script which receives two file names as arguments. It should check whether the two file contents are same or not. If they are same then second file should be deleted.
9. a)Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.

b)Develop an interactive script that ask for a word and a file name and then tells how many times that word occurred in the file. c)Write a shell script to perform the following string operations: i)To extract a sub-string from a given string. ii)To find the length of a given string.
10 .Write a C program that takes one or more file or directory names as command line input and reports the following information on the file: i)File type ii)Number of links iii)Read, write and execute permissions iv)Time of last access (Note : Use stat/fstat system calls)
11. Write C programs that simulate the following unix commands: a)mv b)cp (Use system calls)
12. Write a C program that simulates ls Command (Use system calls / directory API)
13. Write a shell script to accept the name of the file from standard input and perform the following tests on it a) File executable b) File readable c) File writable d) Both readable & writable
14. Write an awk program to print sum, avg of students marks list
15. Write an awk program which will find maximum word and its length in the given input File
Reference Books:
1. Unix & Shell programming – A text book, B.A.Forouzan&R.F.Giberg, Thomson.
2. Unix & Shell programming by YashavantKanetkar

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2206	PC	--	--	3	1.5	15	35	3Hrs.

JAVA PROGRAMMING LAB

(COMMON TO AIDS &IT)

Course Objectives: The aim of this lab is to

1.	Practice programming in theJava
2.	Gain knowledge of object-oriented paradigm in the Java programminglanguage
3.	Learn use of Java in a variety of technologies and on differentplatforms

Course Outcomes: By the end of the course student will be able to write java program for

S.No	Outcome	Knowledge Level
1.	Apply primitive data types, Operations, Expressions, Control-flow,Strings in java programming	K3
2.	Examine Class, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handlingmechanism	K4
3.	Analyzing simple inheritance, multi-level inheritance, Exception handlingmechanism	K4
4.	Analyze and Construct Threads, Event Handling, implement packages	K4

LIST OF EXPERIMENTS

Exercise - 1 (Basics)

1.	Write a JAVA program to display default value of all primitive data type ofJAVA
2.	Write a java program that display the roots of a quadratic equation $ax^2+bx+c=0$. Calculate the discriminate D and basing on value of D, describe the nature ofroot.
3.	Five Bikers Compete in a race such that they drive at a constant speed which may or may not be the same as the other. To qualify the race, the speed of a racer must be more than the average speed of all 5 racers. Take as input the speed of each racer and print back the speed of qualifying racers.

Exercise - 2 (Operations, Expressions, Control-flow, Strings)

1.	Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
2.	Write a JAVA program to sort for an element in a given list of elements using bubblesort
3.	Write a JAVA program to sort for an element in a given list of elements using mergesort.
4.	Write a JAVA program using StringBuffer to delete, removecharacter.

Exercise - 3 (Class, Objects)

1.	Write a JAVA program to implement class mechanism. – Create a class, methods and invoke them inside main method.
2.	Write a JAVA program to implement constructor.

Exercise - 4 (Methods)

1.	Write a JAVA program to implement constructor overloading.
2.	Write a JAVA program implement method overloading.

Exercise - 5 (Inheritance)	
1.	Write a JAVA program to implement Single Inheritance
2.	Write a JAVA program to implement multi level Inheritance
3.	Write a java program for abstract class to find areas of different shapes
Exercise - 6 (Inheritance - Continued)	
1.	Write a JAVA program give example for “super”keyword.
2.	Write a JAVA program to implement Interface. What kind of Inheritance can beachieved?
Exercise - 7 (Exception)	
1.	Write a JAVA program that describes exception handlingmechanism
2.	Write a JAVA program Illustrating Multiple catchclauses
Exercise – 8 (Runtime Polymorphism)	
1.	Write a JAVA program that implements Runtimepolymorphism
2.	Write a Case study on run time polymorphism, inheritance that implements in aboveproblem
Exercise – 9 (User defined Exception)	
1.	Write a JAVA program for creation of Illustratingthrow
2.	Write a JAVA program for creation of Illustratingfinally
3.	Write a JAVA program for creation of Java Built-inExceptions
4.	Write a JAVA program for creation of User DefinedException
Exercise – 10 (Threads)	
1.	Write a JAVA program that creates threads by extending Thread class .First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds ,(Repeat the same by implementingRunnable)
2.	Write a program illustrating **isAlive**and **join()**
Exercise – 11 (Packages)	
1.	Write a JAVA program illustrate classpath
2.	Write a case study on including in class path in your os environment of yourpackage.
3.	Write a JAVA program that import and use the defined your package in the previousProblem
Exercise – 12 (Event Handling)	
1.	Write a JAVA program that display the x and y position of the cursor movement usingMouse.
2.	Write a Java program to create radio buttons(male & female) perform event handling to display relevant text when radio button selected and button press is performed.
3.	Write a java program to Demonstrate KeyAdapter classes.
Exercise- 13 (JDBC)	
1.	Write a Java program to establish connection with database and Retrieve values form a table.
2.	Write a java program to establish connection with database and insert values into the table.
Reference Books:	
1.	JAVA one step ahead, Anitha Seth, B.L.Juneja,Oxford.
2.	The complete Reference Java, 8th edition, Herbert Schildt,TMH.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2207	**SOC**	**-**	**--**	**4**	**2**	**--**	**50**	**3 Hrs**

ANIMATIONS
(Skill Oriented Course-II)
(Common to AIDS, CSBS & IT)

Course Objectives: The objectives of the course are to impart:

1.	This Course will enable students to learn various aspects of animation using a variety of 2-D software and to implement advance principles of traditional animation in Adobe animate to create high quality animation for production.

Course outcomes : After completion of the course, students will be able to		**K L**
1	Learn various tools of digital 2-D animation.	K2
2	Understand production pipeline to create 2-D animation.	K3
3	Analyze special effects in animation to bring interest and awe in the scenes and Back grounds.	K3
4	Apply the tools to create 2D animation for films and videos.	K3

SYLLABUS

Adobe Photoshop	Create your visiting card
	Create Title for any forthcoming film
	Digital Matte Paint
	Convert Black and White to Color
	Convert Day mode to Night mode
	Design Image manipulation
	Smooth skin and remove blemishes & scars
	Create a 3D pop-out effect
	Create Textures
	Timeline Animation
Adobe Illustrator	Advertisement
	Digital Illustrations
	Brochure
	Packet Design(Toothpaste packet, Soap cover, any Food product)
	Danglers for display
	Menu cards
	Calendar Design
	Tracing image
	Vehicle Design
	Festival
Adobe In design	Magazine A4 Size
	Newspaper layout design & advertisements – Fine arts
	Special Supplement

	Different categories of Books
	Info-graphics
	Caricatures
Corel DRAW	Create a paper ad for advertising of any commercial agency
	Package Design
	Corporate ID
	Exhibition Layout
	Oblers
Animation	Creating Web Banners in Adobe Flash
	Creating a Logo Animation in Adobe Flash
	Creating Frame by Frame animation
	Draw Cartoon Animation using reference.
	Create Lip Sink to Characters
	Using filters & Special effects
	Create a scene by using Mask layers animation
	E-Learning Lab:
	Student Application form
	Video Controlling
	Audio Controlling
	Start Drag and Stop Drag Actions
	Interactive Keyboard Controls using Flash Action Script.
	Interactive Flash Game.
	Creating Character Animation in After Effects

Reference Books:

1	Adobe Animate CC Classroom Book 2018 Animation, First Edition, Pearson

Code	Category	L	T	P	C	I.M	E.M	Exam
B20IT2208	**SOC**	**-**	**--**	**4**	**2**	**--**	**50**	**3 Hrs**

WEB PAGE DESIGN USING PHP	
(Skill Oriented Course-II)	
(Common to AIDS, CSBS & IT)	
Course Objectives: The objectives of the course are to impart:	
1.	Understand the principles of creating an effective web page.
2.	Understand elements of design with regard to the web.
3.	Learn the language of the web: HTML and CSS.
4.	Develop skills in analyzing the usability of a web site.
5.	Understand how to develop PHP web application with database connectivity.
6.	Learn CSS grid layout.

Course outcomes : After completion of the course, students will be able to		**K L**
1	Apply the principles of creating an effective web page.	K3
2	Apply the elements of design with regard to the web.	K3
3	Create the language of the web: HTML and CSS.	K4
4	Develop skills in analyzing the usability of a web site.	K4
5	Understand how to plan and conduct user research related to web usability.	K2
6	Create CSS grid layout	K4

SYLLABUS

Exercise 1	**Introduction to HTML**
	1.1 What is HTML
	1.2 HTML Documents
	1.3 Basic structure of an HTML document
	1.4 Creating an HTML document
	1.5 Mark up Tags
	1.6 Heading-Paragraphs
	1.7 Line Breaks
	1.8 HTML Tags.
Exercise 2	**Elements of HTML**
	2.1 Introduction to elements of HTML
	2.2 Working with Text
	2.3 Working with Lists, Tables and Frames
	2.4 Working with Hyperlinks, Images and Multimedia
	2.5 Working with Forms and controls.
Exercise 3	**Introduction to Cascading Style Sheets**
	3.1 Concept of CSS
	3.2 Creating Style Sheet
	3.3 CSS Properties
	3.4 CSS Styling(Background, Text Format, Controlling Fonts)
	3.5 Working with block elements and objects

	3.6 Working with Lists and Tables
	3.7 CSS Id and Class
	3.8 Box Model (Introduction, Border properties, Padding Properties, Margin properties)
Exercise 4	4.1 The Basic of JavaScript: Objects,
	4.2 Primitives Operations and Expressions,
	4.3 Screen Output and Keyboard Input,'
	4.4 Object Creation and Modification, Arrays, Functions
	4.5 DHTML: Positioning Moving and Changing Elements
Exercise 5	5.1 Introducing PHP: Creating PHP script,
	5.2 Running PHP script.
	5.3 Using variables, constants, Datatypes, Operators.
	5.4 Conditional statements, Control statements, Arrays ,functions
	5.5 Working with forms and Databases such as MySQL.
	5.6 Develop PHP MySQL CRUD Application

Text Books:	
1.	Web Technologies, Uttam K Roy, Oxford
2.	HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery) 2Ed,Dreamtech Press; Second edition
3.	The Web Warrior Guide to Web Programming, Bai, Ekedahl, Farrelll, Gosselin, Zak,Karparhi, Maclntyre, Morrissey, Cengage
Reference Books:	
1.	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5, Shroff Publishers & Distributers Private Limited - Mumbai; Fourth edition
2.	PHP: The Complete Reference, McGraw Hill Education; Raunakphp study edition